

MakeYourMove Platform – Detailed Assessment Workflow Document

1. Introduction

MakeYourMove is a behavior-driven career exploration platform designed for software engineering students (1st–4th year) and early-career professionals who are uncertain about which technical specialization aligns with their thinking style.

Unlike traditional career guidance tools that rely on static questionnaires or personality tests, this platform evaluates users through **interactive micro-activities, scenario-based tasks, and behavioral signals**. The system captures how users interact with engineering-like problems and maps those behaviors to potential software engineering domains.

The assessment process consists of three major stages:

1. **Pre-Assessment (Behavioral Screening)**
2. **Domain Exploration Assessment**
3. **Result Visualization and Exploration Guidance**

This document explains the workflow and logic of each stage in detail.

2. Overall System Workflow

The assessment follows a progressive discovery model:

1. **User completes a short behavioral pre-assessment**
2. The system analyzes interaction patterns
3. Domains are **shortlisted into top clusters**
4. The user explores **domain-specific micro activities**
5. The system generates **visual domain alignment results**
6. The user receives **learning paths and curated resources**

Workflow summary:

User → Pre-Assessment → Behavioral Scoring → Top Domain Clusters → Domain Exploration Tasks → Result Visualization → Learning Roadmap + Resources

3. Pre-Assessment Stage

3.1 Objective of the Pre-Assessment

The pre-assessment is designed to:

- Capture **initial behavioral signals**
- Detect **engineering thinking patterns**
- Reduce the full domain space into **3 promising clusters**

- Avoid overwhelming the user with irrelevant activities

This stage **does not produce final recommendations**.

Its goal is only **domain narrowing**.

3.2 Domains Considered by the Platform

The platform initially considers seven major software engineering domains:

- Backend Engineering
- Frontend Development
- DevOps / Infrastructure
- Data Engineering
- Artificial Intelligence / Machine Learning
- Cybersecurity
- Product Engineering

During the pre-assessment, these are compressed into broader **domain clusters**.

Cluster Mapping

Systems & Infrastructure

→ Backend + DevOps

User Experience & Product

→ Frontend + Product Engineering

Data & Intelligence

→ Data Engineering + AI/ML

Security & Reliability

→ Cybersecurity

Hybrid / Balanced

→ Mixed signals

The system ultimately selects **top 3 clusters** for deeper exploration.

4. Pre-Assessment Micro-Activities

The pre-assessment contains **4 short interactive tasks** designed to simulate engineering thinking rather than asking direct questions.

Each activity takes **20–30 seconds**, ensuring the total pre-assessment remains under **3 minutes**.

4.1 Activity 1 – System Flow Arrangement

Objective

Detect whether the user thinks in terms of **system architecture, UI flow, or data processing**.

Interaction

Users see several blocks representing system components:

- User Request
- API
- Database
- Machine Learning Model
- UI Response

Users must **drag and arrange these blocks** to form a logical application flow.

Behavioral Signals Captured

- Ordering logic
- Rearrangement attempts
- Time to completion
- Component selection preference

Domain Signals

API and Database prioritization → Backend inclination

Model inclusion → AI / Data inclination

UI emphasis → Frontend inclination

4.2 Activity 2 – Interface Improvement Task

Objective

Detect **UI thinking, usability awareness, and frontend intuition**.

Interaction

Users see a poorly designed form layout containing:

- Title
- Input fields
- Error messages
- Submit button

The elements are misaligned and confusing.

Users must **drag and reposition components** to improve the interface.

Behavioral Signals

- Logical grouping
- Error placement
- Visual hierarchy decisions

Domain Signals

Clear hierarchy improvements → Frontend / UX

Field-error pairing → Product thinking

4.3 Activity 3 – Debug Investigation Task

Objective

Detect **debugging instincts and system investigation strategy**.

Scenario

The application page is loading slowly.

Users must select **where they would investigate first**.

Available options:

- Database queries
- Network requests
- UI rendering
- Server logs
- Deployment pipeline

Behavioral Signals

- Investigation order
- Selection priorities

Domain Signals

Database focus → Backend

Pipeline logs → DevOps

UI rendering → Frontend

4.4 Activity 4 – Solution Stack Builder

Objective

Evaluate **architecture thinking and infrastructure awareness**.

Scenario

The system must support **1 million users reliably**.

Users select components from:

- Load balancer
- Database
- Redis cache
- Monitoring tools
- Docker containers
- ML model
- Firewall

Users drag components into a **solution architecture area**.

Behavioral Signals

- Component selection
- Architecture design logic
- Infrastructure awareness

Domain Signals

Load balancing / monitoring → DevOps

Caching / database → Backend

ML components → AI

Firewall → Security

5. Behavioral Signal Processing

All user actions are recorded as structured behavioral signals.

Captured data includes:

- Task choices
- Decision order
- Time taken per activity
- Retry attempts
- Interaction patterns

These signals are stored and analyzed using a **rule-based scoring engine**.

6. Cluster Selection Logic

After scoring individual domain signals, the system aggregates scores into clusters.

Example scoring aggregation:

Systems = Backend + DevOps
UX/Product = Frontend + Product
Data Intelligence = Data + AI
Security = Security

The system selects **top 3 clusters** based on relative score strength.

These clusters proceed to the next stage.

7. Result Visualization (Cluster Level)

Instead of percentages, results are displayed using **relative size bubbles**.

This creates an exploratory experience rather than a deterministic label.

Example visualization:

Large bubble → strongest behavioral alignment

Medium bubble → secondary alignment

Small bubble → exploratory domain

Example:

Systems Engineering

Data & Intelligence

User Experience

Accompanying message:

"Based on your interaction patterns, these domains may match your engineering thinking style.
Explore them further through short practical activities."

8. Domain Exploration Stage

Each bubble opens a **domain exploration page** containing three components:

1. Exploration Micro-Activity
2. Learning Roadmap
3. Curated Free Resources

This stage encourages **active experimentation** before committing to a domain.

9. Example Domain Exploration – Data / AI

9.1 Exploration Micro-Activity

Users are asked to perform a small real-world task using **Weka**.

Task Instructions

1. Download Weka

2. Open the Iris dataset
 3. Run a classification algorithm
 4. Observe model accuracy and confusion matrix
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9.2 Verification Quiz

To confirm the activity was performed:

Example questions:

Which algorithm did you run?

Decision Tree / KNN / Naive Bayes

What accuracy range did you observe?

60-70 / 70-80 / 90-100

What does the confusion matrix represent?

Prediction accuracy breakdown

9.3 Learning Roadmap

A staged learning path is provided.

Stage 1 – Programming & statistics fundamentals

Stage 2 – Data cleaning and transformation

Stage 3 – Machine learning models

Stage 4 – Model deployment and pipelines

9.4 Free Learning Resources

Examples:

Books

Hands-On Machine Learning

YouTube

StatQuest

Andrew Ng ML lectures

Practice

Kaggle beginner datasets

10. Example Domain Exploration – Backend

Micro-Activity

Users interact with a REST API using **Postman**.

Task:

1. Send a GET request to a public API
2. Inspect the JSON response
3. Modify parameters

Verification quiz:

Which HTTP method retrieves data?

GET / POST

What format was the response?

JSON / XML

Learning Roadmap

Stage 1 – Programming fundamentals

Stage 2 – REST API design

Stage 3 – Database management

Stage 4 – Scalable backend systems

11. Final Result Generation

After users interact with domain activities, the system updates domain confidence scores.

The final visualization displays **domain bubbles with relative size**.

Example:

Backend Engineering

Data Engineering

DevOps Infrastructure

Bubble size reflects **behavioral alignment strength**.

12. Final Guidance Section

Below the visualization the system displays:

Based on your interaction patterns and exploration tasks, these domains appear most aligned with your engineering approach. Continue exploring the recommended learning paths to validate your interest.

13. Key Design Principles

The platform is built on three principles:

Behavior over self-reporting

Users demonstrate skills through interaction rather than answering preference questions.

Exploration over labeling

The system suggests domains but encourages experimentation.

Actionable learning

Every recommendation is paired with practical tasks and curated resources.

14. Expected Impact

The platform aims to help software engineering students:

- Reduce confusion around specialization choices
- Discover domains through practical interaction
- Develop structured learning paths
- Build confidence in career direction

By combining behavioral analysis with domain exploration, MakeYourMove transforms career guidance into an **interactive discovery experience rather than a static assessment tool**.